

Exercises 09/11

1. Show that $\mathbb{R}_+^2$ is a metric space – be sure to define the metric.
2. Show that if $\{x_n\}$ is a sequence in a metric space (X, d) and if $x_n \rightarrow x$ and $x_n \rightarrow y$ then $x = y$.
3. Let (X, d) be a complete metric space and $\{x_n\}$ a Cauchy sequence in X , so that there is a point $x \in X$ such that $x_n \rightarrow x$. Show this point is unique.
4. Show that if $\{x_n\}$ is a sequence in (X, d) which converges to $x \in X$ then every subsequence of $\{x_n\}$ also converges to x .
5. Let $\{x_n\}$ be a sequence of real numbers.
 - a. Show that $\{x_n\}$ converges to some $x \in \mathbb{R}$ if and only if $\limsup x_n = \liminf x_n$.
 - b. Using the completeness of $\mathbb{R}$ (every non-empty set of real numbers that is bounded above has a least upper bound) and the previous part of this exercise, show that Cauchy sequences converge.
6. Answer the following.
 - a. Show that $\overline{B}(x, \varepsilon) = \{y \in X : d(x, y) \leq \varepsilon\}$ is closed.
 - b. Show that $(1, 2) \cup (3, \infty)$ is open in $\mathbb{R}$.
 - c. Show that the intersection of two open sets is open and of two closed sets is closed.
 - d. Are there any subsets of $\mathbb{R}$ that are both open and closed?